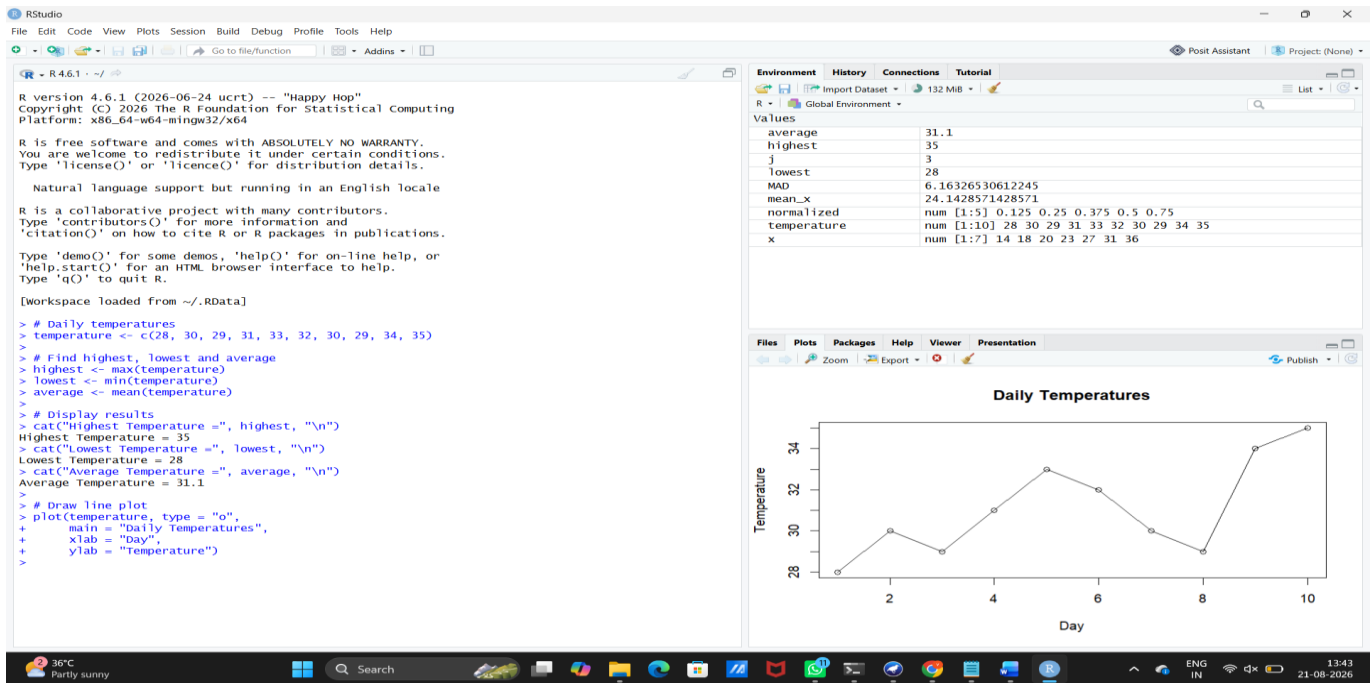
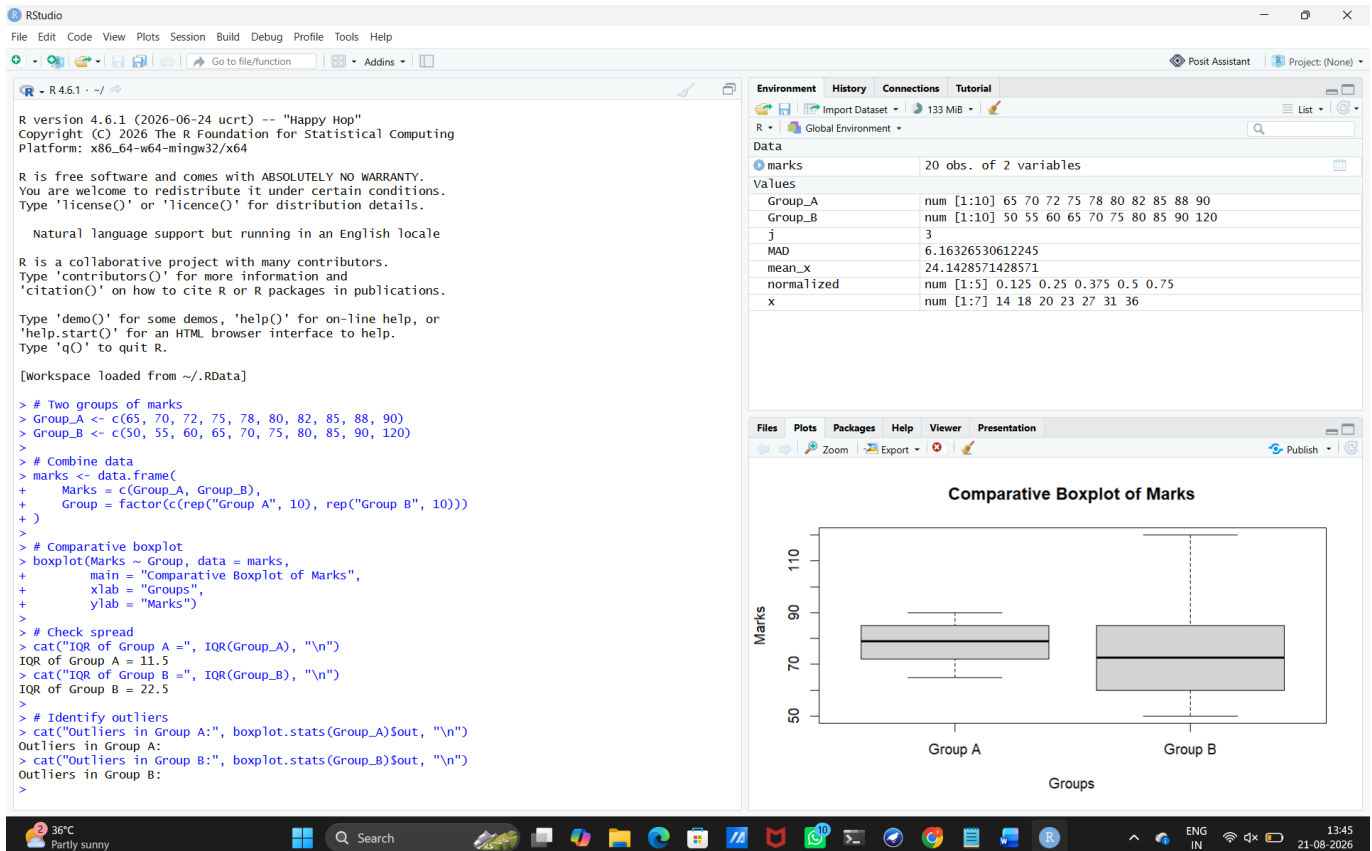


SET 8

1. R: Create 10 daily temperatures; find highest, lowest and average and draw a line plot.



2. R: Draw comparative boxplots for two groups of marks; identify greater spread and outliers.



3. WEKA: Create banking ARFF data (Balance, Age, Transactions, RiskLevel); apply J48 with 10-fold CV.

Dataset: Bank Risk

```
@relation bank_risk
@attribute Balance numeric
@attribute Age numeric
@attribute Transactions numeric
@attribute RiskLevel {Low,Medium,High}
@data
12000,23,8,High
25000,31,15,Medium
45000,42,20,Low
18000,27,10,High
52000,48,25,Low
32000,36,18,Medium
15000,25,7,High
60000,52,28,Low
```

The screenshot shows the Weka Explorer application window. The 'Classify' tab is selected. In the 'Classifier' section, 'J48 -C 0.25 -M 2' is chosen. Under 'Test options', 'Cross-validation' is selected with 'Folds' set to 10. The 'Result list' on the left shows '14:05:47 - trees.J48' selected. The 'Classifier output' pane on the right displays the following information:

```
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    bank_risk
Instances:   8
Attributes:  4
             Balance
             Age
             Transactions
             RiskLevel
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----

Balance <= 18000: High (3.0)
Balance > 18000
|   Balance <= 32000: Medium (2.0)
|   Balance > 32000: Low (3.0)

Number of Leaves :    3

Size of the tree :    5

Time taken to build model: 0 seconds
```

The status bar at the bottom indicates 'Problem evaluating classifier'.

4. WEKA: Apply Naive Bayes to the banking data and compare correct percentage and F1 with J48.

Dataset: Bank Risk – Naïve Bayes

Use the same Bank Risk ARFF dataset from Set 8 Q3.

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

NaiveBayes

Test options

Use training set

Supplied test set

Set...

Cross-validation

Folds

10

Percentage split

%

66

More options...

(Nom) RiskLevel

Start

Stop

Result list (right-click for options)

14:05:47 - treesJ48

14:07:48 - bayes.NaiveBayes

Classifier output

RiskLevel

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute	Class		
	Low (0.36)	Medium (0.27)	High (0.36)
=====			
Balance			
mean	54857.1429	30857.1429	16000
std. dev.	5598.8337	3428.5714	3232.4881
weight sum	3	2	3
precision	6857.1429	6857.1429	6857.1429
Age			
mean	48.3333	33.1429	26.2381
std. dev.	5.1671	4.1429	1.953
weight sum	3	2	3
precision	4.1429	4.1429	4.1429
Transactions			
mean	24	16.5	8
std. dev.	2.4495	1.5	1.4142
weight sum	3	2	3
precision	3	3	3
Time taken to build model: 0 seconds			

Status

Problem evaluating classifier

Log

x 0